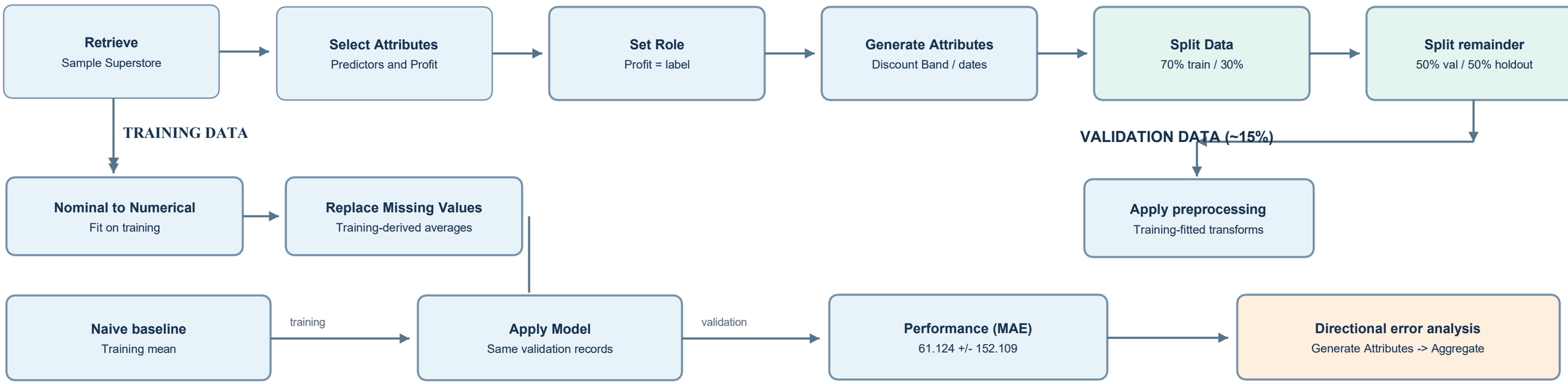
Annotated workflow | Complete process architecture

All five models, shared validation field, prediction/error-analysis branches and selection



Holdout (~15%) remains outside tournament tuning. See page 2 for branch semantics and business cost.

Workflow annotations | How the process executes

Detailed reading guide and final selection evidence

1 | Source and feature

Retrieve Sample Superstore. Select the intended predictors and assign Profit as the numeric label.

Generate Discount Band, Order Year, Order Month, and Order Quarter before model comparison.

2 | Partitioning and leakage

Seed 6729. First split: 70% training and 30% remaining. Second split: half validation, half

Nominal to Numerical and Replace Missing Values are fit on training data; apply learned transforms to

3 | Five parallel regression

Default Model predicts training mean; Linear Regression uses bias and ridge 0.

Decision Tree depth 15; Random Forest 200 trees, depth 10; Neural Network 200 cycles, LR 0.01, momentum 0.9.

4 | Same-field model evaluation

Apply each fitted model to the same validation examples; Performance computes absolute_error (MAE).

Selected tree: 54.223; NN: 54.969; forest: 55.515; baseline: 61.124; linear regression: 95.303.

5 | Directional error

Branch the labeled predictions via Multiply, preserving the original Performance

Generate under_error=max(Profit-prediction,0), over_error=max(prediction-Profit,0); Aggregate averages

6 | Business selection and

Planning cost = 1.50*mean under_error + 1.00*mean over_error + 2.00 per decision.

Decision Tree estimated cost 68.945 per decision. Cost assumptions are illustrative, not verified finance data.

Final leaderboard | Validation and directional errors

Use this plain table alongside the native .rmp process and the documented dataset access path

Model	Validation MAE (+/- spread)	Avg under	Avg over	Expected cost / decision
Decision Tree (depth 15)	54.223 +/- 135.737	25.444	28.779	68.945
Neural Network	54.969 +/- 133.446	30.742	24.227	72.340
Random Forest (200 trees)	55.515 +/- 145.582	26.683	28.832	70.857
Naive Baseline	61.124 +/- 152.109	29.170	31.954	77.709
Linear Regression	95.303 +/- 214.290	56.177	39.126	125.392

Cost assumptions: 1.50 per underprediction unit; 1.00 per overprediction unit; 2.00 fixed cost per decision.

Selection: Decision Tree (depth 15), subject to untouched holdout testing and review of business cost assumptions.

The workflow illustration is a conceptual map of the saved process, not a pixel-identical screenshot.

Workflow annotations and reproducibility notes

1. Data and target. . Retrieve Sample Superstore. Choose features for project and label numeric Profit. Remove Sales and any columns with identifiers/names from predictors. Create Discount Band, Order Year, Order Month and Order Quarter.

2. Partition. Divide dataset 70% training and 30% remainder, split remainder 50/50. Seed 6729. Use the validation part for tournament comparison, holdout for final evaluation.

3. Preprocessing. Nominal to Numerical transforms categorical predictors into numeric indicator variables. Replace Missing Values imputes missing values in numeric predictors. Apply to Validation Data uses the preprocessing model learned from training data on validation data to check the native process for parameter settings.

4. Model tournament. Train the mean Default Model, Linear Regression, Decision Tree, Random Forest and Neural Network on the same training field. Score each model on the same validation examples. Performance is absolute_error (MAE).

5. Segment 7 error branches. After Apply Model, multiply preserves the Performance branch and passes another down to Generate Attributes. Overerror = max(prediction(Profit) Profit, 0); undererror = max(Profit prediction(Profit), 0). Reduction takes the average of both.

6. Business costs. Planning assumptions 1.50 per underprediction unit, 1.00 per overprediction unit and 2.00 operating cost per prediction. E.g. No. of units of cost = 1.50 x mean undererror + 1.00 x mean overerror + 2.00 Units of cost (all assumptions, not audited business costs).

7. Model selection. The Decision Tree at depth 15 has the lowest observed validation MAE and also the lowest expected cost of the planning assumptions. This is a validation-based recommendation and does not imply unseen holdout performance.

Model	Validation MAE	Avg under	Avg over	Expected cost / decision
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